

ECO5050 AI Applications and Techniques in Economics

Lecture 6 Introduction to Machine Learning

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Machine Learning in Economics

- Machine Learning (ML) is typically used to describe a set of algorithmic approaches to statistical learning.
 - ▶ Breiman (2001) “There are two cultures in the use of statistical modeling to reach conclusions from data. One assumes that the data are generated by a given stochastic data model. The other uses algorithmic models and treats the data mechanism as unknown.”
- Differences in the goals of traditional econometrics and ML (Athey and Imbens, 2019)
 - ▶ Econometrics: Specify a parameter of interest, which is estimated by finding the values that best fit the full sample, using an objective function. The focus is on the quality of the estimator, traditionally measured through large sample efficiency.
 - ▶ ML: The focus is typically developing algorithms, to make predictions about some variables given others or to classify units on the basis of limited information

Econometrics and Machine Learning

- For econometrics, ML can be thought of as highly nonparametric.
- Research in Economics emphasize the use of methods with formal properties of a type that many ML methods do not naturally deliver.
 - ▶ Large sample properties of estimators and tests, including consistency, normality and efficiency.
- Difference in terminology

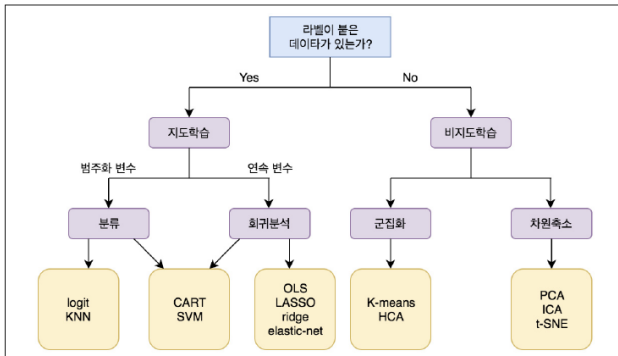
Econometrics	Machine Learning
Independent variables, Predictors	Features
Dependent variable	Target, Label
Sample used to estimate the parameters	Training set
Out-of-sample	Test set

Types of Machine Learning

- Supervised learning
 - ▶ The training data you feed into the algorithm includes the desired solution (labels Y)
 - ▶ Commonly used for prediction
- Unsupervised learning
 - ▶ The training data is unlabeled
 - ▶ Used for clustering, dimensionality reduction, anomaly detection
- Reinforcement learning
 - ▶ The learning system (agent) can observe the environment, select and perform actions, and get rewards in return.
 - ▶ Dynamic optimization based on a Markov process

Types of Machine Learning

〈그림 1〉 지도 학습과 비지도 학습의 구분(Supervised learning and unsupervised learning)



박기영, 고정원 (2019)

Machine Learning in Economics

- Recent literature suggests that ML could add value to economic research and analysis by (Desai, 2023)
 - ▶ Processing non-traditional data, such as images, texts, audios and videos
 - ▶ Capturing non-linearity that is difficult to capture using traditional models
 - ▶ Processing traditional data at scale to improve the accuracy of predictions, extract new information or automate feature extraction

Plans

- Supervised ML for prediction
 - ▶ Regularized linear regression
 - ▶ Regression trees
 - ▶ Ensemble methods: Random Forests, Boosting
 - ▶ Neural Network: Recurrent Neural Network (RNN), Long Short-Term Memory (LSTM), Convolutional Neural Network (CNN)
- Other applications
 - ▶ Causal ML: Athey and Imbens (2016), Wager and Athey (2018)
 - ▶ Interpretation of ML predictions: Shapley additive explanations (Lundberg and Lee, 2017; Strumbelj and Kononenko, 2010)

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